



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 • STANDARD 6.AT.7

# Identify Equivalent Algebraic Expressions

Lesson 6-6 • Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_



## TODAY'S OBJECTIVES

**Content Objective:** I can show that two expressions are equivalent by simplifying and combining like terms.

**Language Objective:** I can explain my work using the words equivalent, simplify, like terms, and combine.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



**Fill in each blank as we go. Use the Word Bank to help you.**



## WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Equivalent Expressions

Equivalent

Simplify

Like Terms

Combine

Coefficient



Tap any word to see its meaning and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

Expressions with the same value for  allowed number are equivalent.

2

Expressions that always have the same value are .

3

Rewriting an expression in a shorter form that is still equal to it is to  it.

4

Terms with the same variable raised to the same power, like  $4x$  and  $9x$ , are .

5 To add or subtract like terms into one term is to  
them.

6 A number multiplied by a variable is a  .

## Write About the Math

**Are these expressions equivalent? How can you prove it?**

*¿Estas expresiones son equivalentes? ¿Cómo lo puedes probar?*

Say your answer to a partner first. Then write it, using at least two of these words:

☐

**Equivalent Expressions**

*Expresiones equivalentes*

☐

**Equivalent**

*Equivalente*

☐

**Simplify**

*Simplificar*

☐

**Like Terms**

*Términos semejantes*

☐

**Combine**

*Combinar*

**Model:**  $3x + 2x$  combines to  $5x$ , so the two forms are equivalent. — Students explain  $3x + 2x = 5x$  (combining like terms), so both equal  $5x + 10$ , and verify by substituting a value like  $x = 4$  to get 30 both ways.

### Start

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- The expressions \_\_\_\_ equivalent because when I simplify  $4h + 2h + 30$ , I get \_\_\_\_ . I can prove it by substituting  $h =$  \_\_\_\_ : both expressions equal \_\_\_\_ .
- My answer is \_\_\_\_ because \_\_\_\_ .

*Mi respuesta es \_\_\_\_ porque \_\_\_\_ .*

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### Build

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- My claim is \_\_\_\_ .  
*Mi afirmación es \_\_\_\_ .*
- My evidence is \_\_\_\_ , so \_\_\_\_ .  
*Mi evidencia es \_\_\_\_ , entonces \_\_\_\_ .*

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### Explain

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

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☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

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## Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

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## Answer Key & Teacher Guide

1. **Notes 1:** Equivalent Expressions
2. **Notes 2:** Equivalent
3. **Notes 3:** Simplify
4. **Notes 4:** Like Terms
5. **Notes 5:** Combine
6. **Notes 6:** Coefficient
7. **Watch for this mistake:** *A common mistake in Equivalent Expressions is adding the exponents instead of the coefficients when combining like terms — writing  $3x + 2x$  as  $5x^2$  instead of  $5x$ . Remember:  $3x$  means 3 groups of  $x$  and  $2x$  means 2 more groups of  $x$ , so you add the coefficients ( $3 + 2 = 5$ ) and keep the variable's power the same ( $x^1$ , not  $x^2$ ).*
8. **Try It 1:** C.  $4x - 3x + x$  is 3 groups of  $x$  plus 1 more group of  $x = 4x$ . Add the coefficients:  $3 + 1 = 4$ . Check  $x = 3$ :  $3(3) + 3 = 12$  and  $4(3) = 12$ .
9. **Try It 2:** A.  $2x + 5$  — Order of addition does not change the value (commutative property):  $5 + 2x = 2x + 5$ . Check  $x = 10$ :  $5 + 2(10) = 25$  and  $2(10) + 5 = 25$ .